

Student Number	
Mark / 48	

Chemistry

The Chemical Earth and Metals
Modules Test • 2004

General Instructions

- Reading time – 5 minutes
- Working time – 55 minutes
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper and may be removed for convenience
- Write your Student Number at the top of this page

Total Marks – 48

Part A – 12 marks

- Attempt Questions 1 – 12
- Allow about 10 minutes for this part

Part B – 36 marks

- Attempt Questions 13 – 28
- Allow about 45 minutes for this part

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Part A – 12 marks
Attempt Questions 1–12
Allow about 10 minutes for this part

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct ↖

Answer Box for Questions 1- 12				
1	A ☐	B ☐	C ☐	D ☐
2	A ☐	B ☐	C ☐	D ☐
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11	A ☐	B ☐	C ☐	D ☐
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1 Which element is found uncombined in the biosphere?

- (A) Carbon
- (B) Hydrogen
- (C) Oxygen
- (D) Neon

2 Which of the following elements has the smallest first ionisation energy?

- (A) Lithium
- (B) Sodium
- (C) Fluorine
- (D) Chlorine

3 The properties of elements X, M, Z and T were investigated and the results tabled...

Element	Malleability	Hardness	Electrical Conductivity (liquid state)
X	malleable	hard	high
M	brittle	soft	nil
Z	brittle	soft	high
T	brittle	hard	low

Choose the correct classification of elements X, M, Z and T based on the data.

	X	M	Z	T
(A)	metal	non-metal	carbon	semi-metal
(B)	non-metal	non-metal	carbon	metal
(C)	semi-metal	metal	carbon	semi-metal
(D)	metal	semi-metal	carbon	metal

4 Which of these substances does not exist as a molecule?

- (A) $C_6H_{12}O_6$
- (B) H_2
- (C) He
- (D) NaCl

- 5 Which of these compounds has the highest melting point?
- (A) LiH
 - (B) SiH₄
 - (C) NH₃
 - (D) HCl
- 6 Which of the following is an application of gravimetric analysis?
- (A) Determining the melting point of a metal alloy.
 - (B) Determining the salinity of the Murray River by testing its electrical conductivity.
 - (C) Testing the flesh of tuna for the presence of mercury.
 - (D) Determining the mass percentage of iron in molasses.
- 7 Which statement is correct for element X located in period 4 and group 7?
- (A) X is a solid that sublimes to a purple vapour.
 - (B) X covalently bonds with sodium to form NaX.
 - (C) X is a liquid at room temperature.
 - (D) X has a higher electronegativity than oxygen.
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- 9 Lithium and sodium have similar physical and chemical properties. This is best explained by which of the following statements?
- (A) They are both metals.
 - (B) They have the same outer shell electronic configuration.
 - (C) They have low relative atomic mass.
 - (D) They are in period 1 of the Periodic Table.

- 10 The melting points of four ionic salts are shown in the table...

Ionic salt	Melting point (°C)
Sodium iodide	660
Silver nitrate	209
Rubidium bromide	692
Potassium sulfate	1072

Which list gives the correct order of the strength of bonds between the indicated ions?

- (A) $\text{Ag}^+ - \text{NO}_3^- > \text{Rb}^+ - \text{Br}^- > \text{Na}^+ - \text{I}^- > \text{K}^+ - \text{SO}_4^{2-}$
(B) $\text{K}^+ - \text{SO}_4^{2-} > \text{Rb}^+ - \text{Br}^- > \text{Na}^+ - \text{I}^- > \text{Ag}^+ - \text{NO}_3^-$
(C) $\text{Ag}^+ - \text{NO}_3^- > \text{Na}^+ - \text{I}^- > \text{Rb}^+ - \text{Br}^- > \text{K}^+ - \text{SO}_4^{2-}$
(D) $\text{K}^+ - \text{SO}_4^{2-} > \text{Na}^+ - \text{I}^- > \text{Rb}^+ - \text{Br}^- > \text{Ag}^+ - \text{NO}_3^-$
- 11 Which of these processes is used in the purification stage of producing pure, 99.96–99.99% copper?
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(D) smelting
- 12 The table shows the properties and uses of four metal alloys...

Alloy	Properties	Uses
W	malleable	car bodies, pipes, nuts and bolts, roofing
X	hard and shock resistant	security safes, files, ball bearings
Y	resists corrosion, easily cast	ships' propellers, casting statues
Z	low melting point, adheres firmly to other metals when molten	joining metals together, applications in plumbing and electronics

Which alloy best describes the properties and uses of solder?

- (A) W
(B) X
(C) Y
(D) Z

Part B – 36 marks

Attempt Questions 13 – 28

Allow about 45 minutes for this part

► *Show all relevant working in questions involving calculations.*

Question 13 (3 marks)

(a) Complete the table showing the existence of compounds on the Earth... **(2 marks)**

	Example of a compound found in the 'sphere
atmosphere	
biosphere	
hydrosphere	
lithosphere	

(b) Explain why uncombined elements are rare. **(1 mark)**

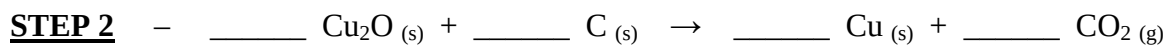
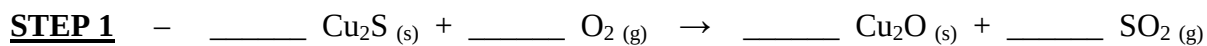
Question 14 (2 marks)

Name these compounds...

Fe_2S_3	
$\text{Fe}(\text{OH})_2$	
NO_2	
N_2O_4	

Question 15 (2 marks)

Chalcocite, a copper sulfide ore is converted into crude copper by a two-step process...



Balance each of the chemical equations. **(2 marks)**

Question 16 (4 marks)

Calcium reacts vigorously with sulfur producing calcium sulfide via an electron transfer reaction.

Complete the table detailing the process...

	Calcium atom	+	Sulfur atom	→	Calcium ion	+	Sulfide ion
Electronic configuration	2-8-8-2	+		→		+	
Lewis electron dot structure		+		→		+	

Question 17 (2 marks)

To separate the components of a mixture, differences in physical properties are used as the basis of the separation. Complete the table to illustrate this principle...

Separation Process	Physical property which enables the separation to occur
distillation	
evaporation	
filtration	
sieving	

Question 18 (2 marks)

For a Prac test, a student is given the task of separating a sample of water collected at Bondi Beach into its three components (water, salt and sand) and determining the mass percentage of each.

A student records this data after filtration and evaporation...

Mass of original sample	226.73 g
Mass of filter paper	1.33 g
Mass of filter paper + dry sand	28.08 g
Mass of salt recovered after evaporation	9.66 g

Calculate the percentage mass of water in the original sample.

Question 19 (3 marks)

Active metal, M, forms M^{2+} when it reacts.

Write balanced formulae equations for the reactions of metal, M, with...

- (i) oxygen
- (ii) water
- (iii) hydrochloric acid

(i) _____

(ii) _____

(iii) _____

Question 20 (2 marks)

Explain why the formula for an ionic compound is an empirical formula.

Question 21 (2 marks)

The reactions of metals with acids requires the transfer of electrons.

Write two half-equations to show the reaction between aluminium and sulfuric acid.

Question 22 (1 mark)

Identify **one** common element that exists as a covalent molecular structure and **one** common element that exists as a covalent network structure.

covalent molecule _____

covalent network _____

Question 23 (2 marks)

Identify and explain the trend in the atomic radius from left to right across the Periodic Table.

Question 24 (3 marks)

The table shows the melting points of the chlorides of seven consecutive elements A – F in a period of the Periodic Table...

Element	A	B	C	D	E	F
Melting point of the element's chloride (°C)	801	712	193	– 68	– 91	– 80

- (a) Identify an element (A – F) which could be a metal. _____
- (b) Identify the type of bonding which exists in the chlorides of elements A and B. _____
- (c) Compare the electrical conductivity of the chlorides, A – F in the liquid state.

Question 25 (2 marks)

Mendeleev's model of the Periodic Table in 1869 was a landmark in the history of Chemistry.

- (a) Describe how Mendeleev used the properties of the known elements to develop his model for the Periodic Table. **(1 mark)**

- (b) Identify one chemical property of iodine which confirms that it is a halogen and should be placed in group 7. **(1 mark)**

Question 26 (2 marks)

In the lab this semester, you decomposed copper(II) carbonate by strongly heating it.
Write a (a) word equation and (b) a balanced formulae equation for the decomposition reaction.

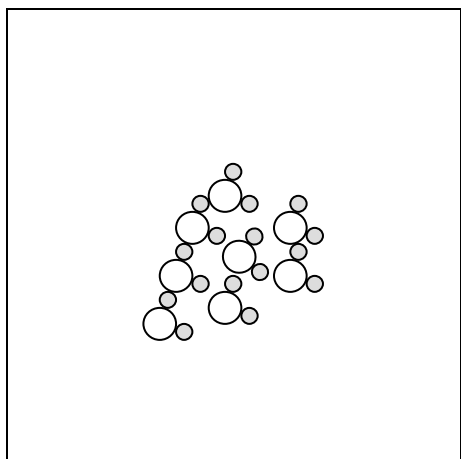
► Assume the reactant is pure copper(II) carbonate.

(a) _____

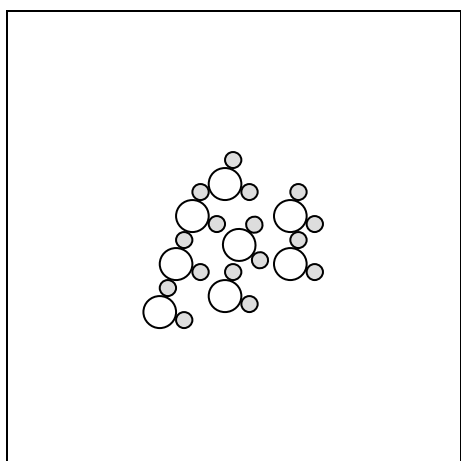
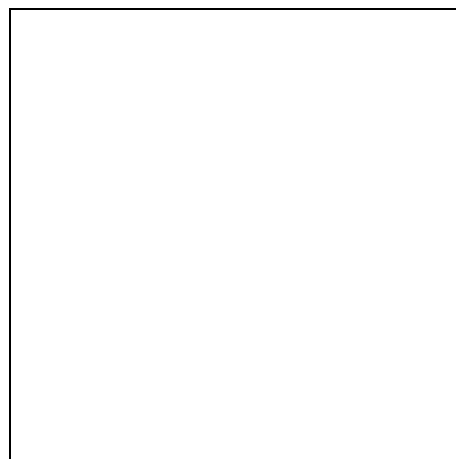
(b) _____

Question 27 (2 marks)

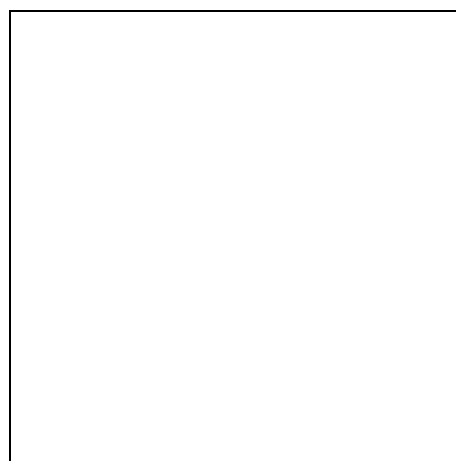
The diagrams on the left show eight water molecules in the liquid state. Complete the diagram boxes on the right, showing the arrangement of particles after all the water molecules have undergone the processes of boiling and electrolysis.



BOILING



ELECTROLYSIS



Question 28 (2 marks)

- (a) Identify one metal and one non-metal. **(1 mark)**

metal _____ *non-metal* _____

- (b) Account for the use of ***either*** the metal ***or*** the non-metal you identified in terms of its ***physical properties***. **(1 mark)**

HIGHER SCHOOL CERTIFICATE EXAMINATION

Chemistry

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+] \qquad \Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

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KEY																	
		Atomic Number		Atomic Weight		Symbol of element		Name of element									
1 H 1.008 Hydrogen	4 Be 9.012 Beryllium	3 Li 6.941 Lithium	79 Au 197.0 Gold	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.39 Zinc	31 Ga 69.72 Gallium	32 Ge 72.61 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton	10 Ne 20.18 Neon	2 He 4.003 Helium	

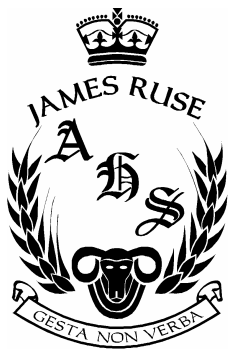
Lanthanides

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [146.9] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinides

89 Ac [227.0] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237.0] Neptunium	94 Pu [239.1] Plutonium	95 Am [241.1] Americium	96 Cm [244.1] Curium	97 Bk [249.1] Berkelium	98 Cf [252.1] Californium	99 Es [252.1] Einsteinium	100 Fm [257.1] Fermium	101 Md [258.1] Mendelevium	102 No [259.1] Nobelium	103 Lr [262.1] Lawrencium
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Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.



Answers and Marking Scheme

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A  B  C  D 

correct (arrow pointing to B)

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(C)	semi-metal	metal	carbon	semi-metal
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Which list gives the correct order of the strength of bonds between the indicated ions?

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(B) $\text{K}^+ - \text{SO}_4^{2-} > \text{Rb}^+ - \text{Br}^- > \text{Na}^+ - \text{I}^- > \text{Ag}^+ - \text{NO}_3^-$
(C) $\text{Ag}^+ - \text{NO}_3^- > \text{Na}^+ - \text{I}^- > \text{Rb}^+ - \text{Br}^- > \text{K}^+ - \text{SO}_4^{2-}$
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Y	resists corrosion, easily cast	ships' propellers, casting statues
Z	low melting point, adheres firmly to other metals when molten	joining metals together, applications in plumbing and electronics

Which alloy best describes the properties and uses of solder?

- (A) W
(B) X
(C) Y
(D) Z

Part B – 36 marks

Attempt Questions 13 – 28

Allow about 45 minutes for this part

► *Show all relevant working in questions involving calculations.*

Question 13 (3 marks)

(a) Complete the table showing the existence of compounds on the Earth... **(2 marks)**

	Example of a compound found in the 'sphere
atmosphere	<i>carbon dioxide</i>
biosphere	<i>glucose</i>
hydrosphere	<i>water</i>
lithosphere	<i>silicon dioxide</i>

(b) Explain why uncombined elements are rare. **(1 mark)**

The vast majority of elements are chemically reactive. They chemically react with other elements to form stable compounds achieving noble gas configurations with lower energy.

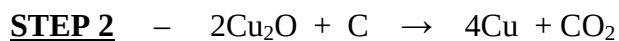
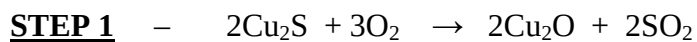
Question 14 (2 marks)

Name these compounds...

Fe_2S_3	<i>iron(III) sulfide</i>
$\text{Fe}(\text{OH})_2$	<i>iron(II) hydroxide</i>
NO_2	<i>nitrogen dioxide</i>
N_2O_4	<i>dinitrogen tetroxide</i>

Question 15 (2 marks)

Chalcocite, a copper sulfide ore is converted into crude copper by a two-step process...



Balance each of the chemical equations. **(2 marks)**

Question 16 (4 marks)

Calcium reacts vigorously with sulfur producing calcium sulfide via an electron transfer reaction.

Complete the table detailing the process...

	Calcium atom	+	Sulfur atom	→	Calcium ion	+	Sulfide ion
Electronic configuration	2-8-8-2	+	2-8-6	→	2-8-8	+	2-8-8
Lewis electron dot structure	$\text{Ca}:$	+	$\begin{array}{c} \cdot \\ :\text{S}: \\ \cdot \end{array}$	→	Ca^{2+}	+	$\left[\begin{array}{c} \cdot\cdot \\ :\text{S}: \\ \cdot\cdot \end{array} \right]^{2-}$

Question 17 (2 marks)

To separate the components of a mixture, differences in physical properties are used as the basis of the separation. Complete the table to illustrate this principle...

Separation Process	Physical property which enables the separation to occur
distillation	<i>boiling point</i>
evaporation	<i>boiling point</i>
filtration	<i>particle size solubility</i>
sieving	<i>particle size</i>

Question 18 (2 marks)

For a Prac test, a student is given the task of separating a sample of water collected at Bondi Beach into its three components (water, salt and sand) and determining the mass percentage of each.

A student records this data after filtration and evaporation...

Mass of original sample	226.73 g
Mass of filter paper	1.33 g
Mass of filter paper + dry sand	28.08 g
Mass of salt recovered after evaporation	9.66 g

Calculate the percentage mass of water in the original sample.

$$\text{Mass of sand} = 28.08 \text{ g} - 1.33 \text{ g} = 26.75 \text{ g} \quad (1 \text{ mark})$$

$$\text{Mass of water} = 226.73 \text{ g (sample)} - 26.75 \text{ g (sand)} - 9.66 \text{ g (salt)} = 190.32 \text{ g}$$

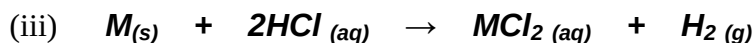
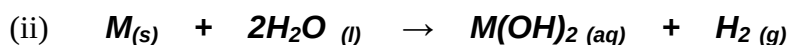
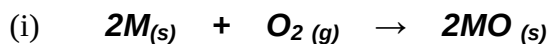
$$\text{Percentage water} = 190.32 \text{ g} \div 226.73 \text{ g} = 83.94\%$$

Question 19 (3 marks)

Active metal, M, forms M^{2+} when it reacts.

Write balanced formulae equations for the reactions of metal, M, with...

- (i) oxygen
- (ii) water
- (iii) hydrochloric acid



Question 20 (2 marks)

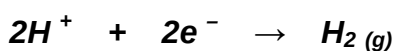
Explain why the formula for an ionic compound is an empirical formula.

The empirical formula represents the simplest whole number ratio of atoms/ions of different elements in a compound. Ionic compounds are an infinite lattice/array of positive and negative ions and therefore the formula given to describe them is just the simplest ratio of ions in the array, i.e. the empirical formula.

Question 21 (2 marks)

The reactions of metals with acids requires the transfer of electrons.

Write two half-equations to show the reaction between aluminium and sulfuric acid.



Question 22 (1 mark)

Identify **one** common element that exists as a covalent molecular structure and **one** common element that exists as a covalent network structure.

covalent molecule **e.g. hydrogen or H_2**

covalent network **e.g. carbon (diamond)**

Question 23 (2 marks)

Identify and explain the trend in the atomic radius from left to right across the Periodic Table.

The atomic radii of elements decrease from left to right (1 mark) across the Periodic Table, as the nuclear charge increases. The greater the nuclear charge, the more the electron shells are attracted to the nucleus and the radius decreases. (1 mark)

Question 24 (3 marks)

The table shows the melting points of the chlorides of seven consecutive elements A – F in a period of the Periodic Table...

Element	A	B	C	D	E	F
Melting point of the element's chloride (°C)	801	712	193	– 68	– 91	– 80

- (a) Identify an element (A – F) which could be a metal. **A, B or C**
- (b) Identify the type of bonding which exists in the chlorides of elements A and B. **ionic**
- (c) Compare the electrical conductivity of the chlorides, A – F in the liquid state.

Chlorides A – C would be electrolytes.

Chlorides D – F would be non-electrolytes.

Question 25 (2 marks)

Mendeleev's model of the Periodic Table in 1869 was a landmark in the history of Chemistry.

- (a) Describe how Mendeleev used the properties of the known elements to develop his model for the Periodic Table. **(1 mark)**

Mendeleev studied the patterns in the elemental data and organised chemically similar elements in columns (groups).

OR

Mendeleev realised that the properties of elements were a periodic function of their atomic weights.

- (b) Identify one chemical property of iodine which confirms that it is a halogen and should be placed in group 7. **(1 mark)**

Iodine (atoms) react in a 1:1 ratio with group one elements to form ionic iodides.

Question 26 (2 marks)

In the lab this semester, you decomposed copper(II) carbonate by strongly heating it. Write a (a) word equation and (b) a balanced formulae equation for the decomposition reaction.

► Assume the reactant is pure copper(II) carbonate.

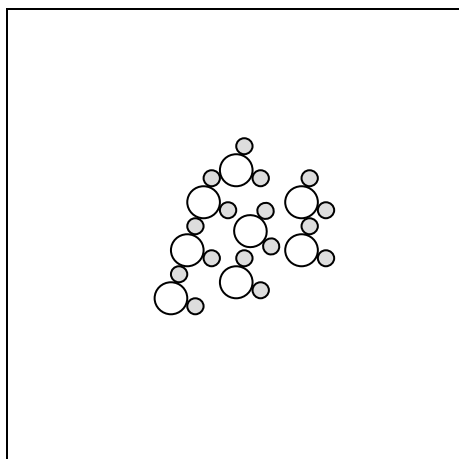
- (a) ***copper(II) carbonate → copper(II) oxide + carbon dioxide***

- (b) ***$\text{CuCO}_3 (\text{s}) \rightarrow \text{CuO} (\text{s}) + \text{CO}_2 (\text{g})$***

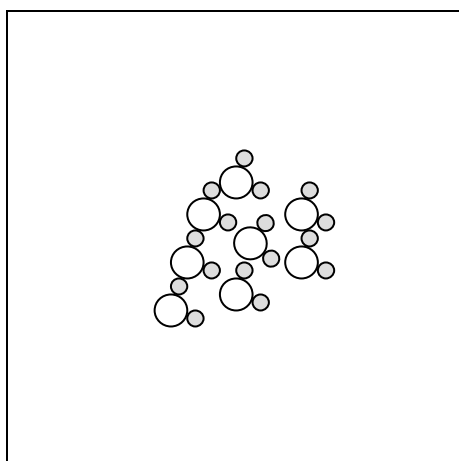
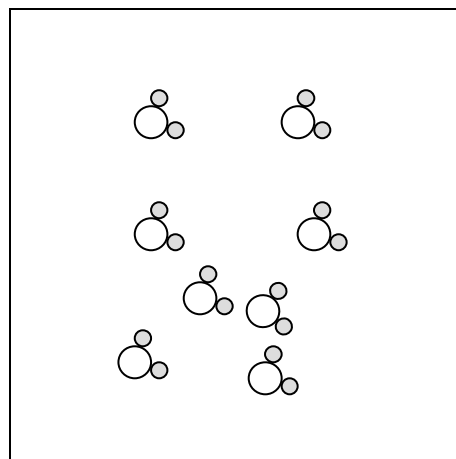
► ***Correct formulae including phase label required for 1 mark.***

Question 27 (2 marks)

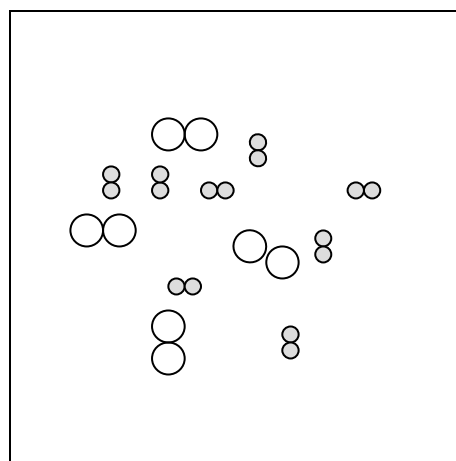
The diagrams on the left show eight water molecules in the liquid state. Complete the diagram boxes on the right, showing the arrangement of particles after all the water molecules have undergone the processes of boiling and electrolysis.



BOILING



ELECTROLYSIS



1 mark each for boiling and electrolysis

► **Mass must be conserved in the transformation.**

Question 28 (2 marks)

- (a) Identify one metal and one non-metal. **(1 mark)**

metal e.g. copper

non-metal e.g. helium

- (b) Account for the use of ***either*** the metal ***or*** the non-metal you identified in terms of its ***physical properties***. **(1 mark)**

Copper is used as electrical wire due to its good electrical conductivity and ductility.

OR

Helium is used as a filler for balloons because of its low density.

HIGHER SCHOOL CERTIFICATE EXAMINATION

Chemistry

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+] \qquad \Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																		2	
1 H 1.008 Hydrogen		KEY																He 4.003 Helium	
		Atomic Number Atomic Weight																	
		79 Au 197.0																	
		Symbol of element Name of element																	
		5 B 10.81 Boron																	
		6 C 12.01 Carbon																	
		7 N 14.01 Nitrogen																	
		8 O 16.00 Oxygen																	
		9 F 19.00 Fluorine																	
		10 Ne 20.18 Neon																	
3 Li 6.941 Lithium		12 Mg 24.31 Magnesium																18 Ar 39.95 Argon	
11 Na 22.99 Sodium		21 Sc 44.96 Scandium																36 Kr 83.80 Krypton	
19 K 39.10 Potassium		22 Ti 47.87 Titanium																37 Rb 85.47 Rubidium	
37 Rb 85.47 Rubidium		23 V 50.94 Vanadium																54 Xe 131.3 Xenon	
85 At [223.0]		24 Cr 52.00 Chromium																86 Rn [222.0]	
Fr [223.0]		25 Mn 54.94 Manganese																87 Fr [223.0]	
		26 Fe 55.85 Iron																	
		27 Co 58.93 Cobalt																	
		28 Ni 58.69 Nickel																	
		29 Cu 63.55 Copper																	
		30 Zn 65.39 Zinc																	
		31 Ga 69.72 Gallium																	
		32 Ge 72.61 Germanium																	
		33 As 74.92 Arsenic																	
		34 Se 78.96 Selenium																	
		35 Br 79.90 Bromine																	
		36 Kr 83.80 Krypton																	
		37 Rb 85.47 Rubidium																	
		38 Sr 87.62 Strontium																	
		39 Y 88.91 Yttrium																	
		40 Zr 91.22 Zirconium																	
		41 Nb 92.91 Niobium																	
		42 Mo 95.94 Molybdenum																	
		43 Tc [98.91] Technetium																	
		44 Ru 101.1 Ruthenium																	
		45 Rh 102.9 Rhodium																	
		46 Pd 106.4 Palladium																	
		47 Ag 107.9 Silver																	
		48 Cd 112.4 Cadmium																	
		49 In 114.8 Indium																	
		50 Sn 118.7 Tin																	
		51 Sb 121.8 Antimony																	
		52 Te 127.6 Tellurium																	
		53 I 126.9 Iodine																	
		54 Xe 131.3 Xenon																	
		55 Cs 132.9 Cesium																	
		56 Ba 137.3 Barium																	
		57-71 Lanthanides																	
		72 Hf 178.5 Hafnium																	
		73 Ta 180.9 Tantalum																	
		74 W 183.8 Tungsten																	
		75 Re 186.2 Rhenium																	
		76 Os 190.2 Osmium																	
		77 Ir 192.2 Iridium																	
		78 Pt 195.1 Platinum																	
		79 Au 197.0 Gold																	
		80 Hg 200.6 Mercury																	
		81 Tl 204.4 Thallium																	
		82 Pb 207.2 Lead																	
		83 Bi 209.0 Bismuth																	
		84 Po [210.0] Polonium																	
		85 At [210.0] Astatine																	
		86 Rn [222.0] Radon																	
		87 Fr [223.0] Francium																	
		88 Ra [226.0] Radium																	
		89-103 Actinides																	
		104 Rf [261.1] Rutherfordium																	
		105 Db [262.1] Dubnium																	
		106 Sg [263.1] Seaborgium																	
		107 Bh [264.1] Bohrium																	
		108 Hs [265.1] Hassium																	
		109 Mt [268] Meitnerium																	
		110 Uun — Ununium																	
		111 Uuu — Ununium																	
		112 Uub — Unubium																	
		113 Uuc — Unucium																	
		114 Uuq — Unuquadium																	
		115 Uut — Ununtrium																	
		116 Uuh — Ununhexium																	
		117 Uus — Ununseptium																	
		118 Uuo — Ununoctium																	

Lanthanides

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm [146.9]	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
Lanthanum	Cerium	Praseodymium	Neodymium	Promethium	Samarium	Europium	Gadolinium	Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium

Actinides

89 Ac [227.0]	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np [237.0]	94 Pu [239.1]	95 Am [241.1]	96 Cm [244.1]	97 Bk [249.1]	98 Cf [252.1]	99 Es [252.1]	100 Fm [257.1]	101 Md [258.1]	102 No [259.1]	103 Lr [262.1]
Actinium	Thorium	Protactinium	Uranium	Nepunium	Plutonium	Americium	Curium	Berkelium	Californium	Einsteinium	Fermium	Mendelevium	Nobelium	Lawrencium

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.